

Minor Project 1

Heart Disease Prediction using Supervised Machine Learning

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Section: B

Course: MCA

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1. Title

Heart Disease Prediction using Supervised Machine Learning Classification Models

2. Problem Statement

Cardiovascular disease is among the leading causes of death globally, and early detection plays a critical role in improving patient outcomes. The objective of this project is to build a supervised machine learning classification system that predicts whether a patient is likely to have heart disease, based on a set of clinical and diagnostic attributes such as age, blood pressure, cholesterol level, chest pain type, and other cardiac indicators. This is framed as a binary classification problem, where the model predicts one of two outcomes: presence or absence of heart disease.

3. Dataset Description

The dataset used is the Heart Disease Prediction dataset, a commonly used dataset for classification tasks, derived from the original UCI Heart Disease (Cleveland) database.

Property	Value
Source	Kaggle (derived from UCI Heart Disease - Cleveland database)
Total Records	270
Total Features	13 input features + 1 target column
Problem Type	Binary Classification
Target Variable	Heart Disease (Presence / Absence)

Feature Description

Feature	Description
Age	Age of the patient (years)
Sex	Sex (1 = male, 0 = female)
Chest pain type	Type of chest pain experienced (1-4)
BP	Resting blood pressure (mm Hg)
Cholesterol	Serum cholesterol (mg/dl)
FBS over 120	Fasting blood sugar > 120 mg/dl (1 = true, 0 = false)
EKG results	Resting electrocardiographic results (0-2)
Max HR	Maximum heart rate achieved
Exercise angina	Exercise induced angina (1 = yes, 0 = no)
ST depression	ST depression induced by exercise relative to rest
Slope of ST	Slope of the peak exercise ST segment
Number of vessels fluoro	Number of major vessels colored by fluoroscopy (0-3)

Feature	Description
Thallium	Thallium stress test result
Heart Disease	Target: Presence / Absence of heart disease

4. Data Preprocessing Steps

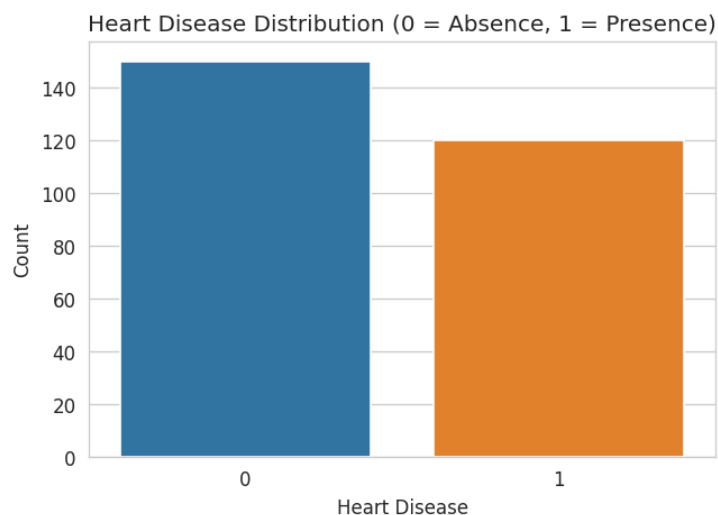
The following preprocessing steps were applied to prepare the dataset for model training:

- **Missing Value Check:** The dataset was inspected for missing/null values; none were found across any of the 14 columns.
- **Duplicate Check:** The dataset was checked for duplicate rows; no duplicates were found, so the full 270 records were retained.
- **Target Encoding:** The target column (Heart Disease) originally contained text values 'Presence' and 'Absence'. These were encoded numerically as 1 (Presence) and 0 (Absence) respectively.
- **Train-Test Split:** The dataset was split into 80% training data (216 records) and 20% testing data (54 records) using stratified sampling to preserve the original class ratio.
- **Feature Scaling:** StandardScaler was applied to normalize all feature values (zero mean, unit variance), essential for distance/margin-based algorithms such as Logistic Regression and SVM.

5. Exploratory Data Analysis

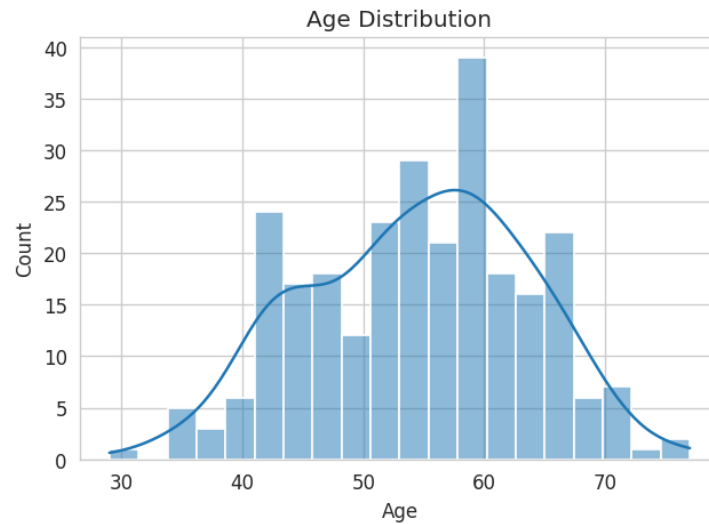
Exploratory Data Analysis (EDA) was performed to understand the structure of the data, the distribution of key variables, and relationships between features and the target class.

5.1 Target Class Distribution



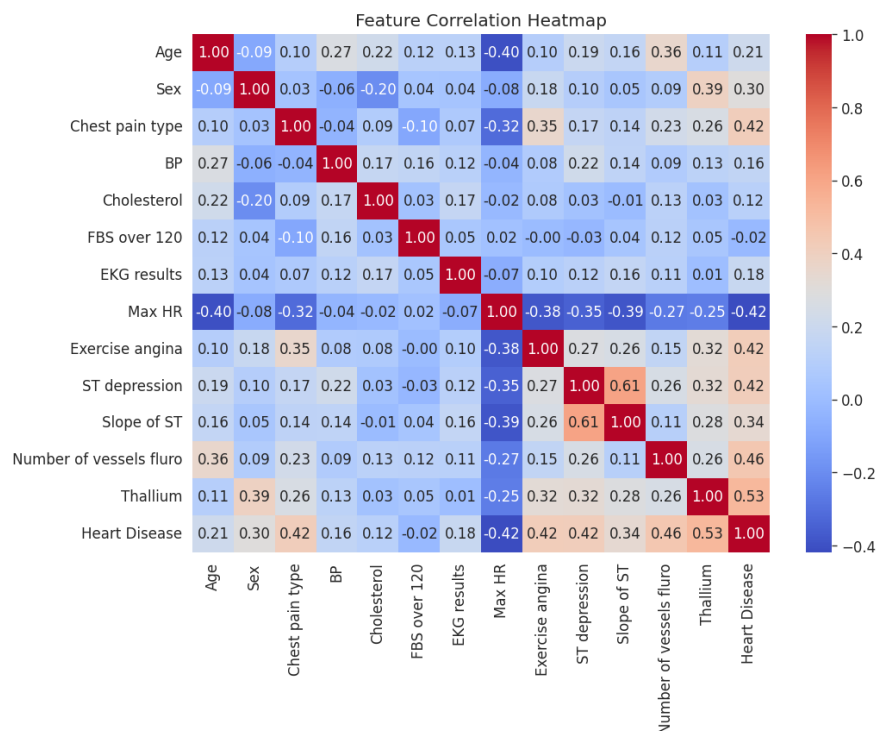
Out of 270 patients, 150 do not have heart disease (Absence) and 120 have heart disease (Presence). The classes are reasonably balanced, meaning accuracy can be treated as a reliable evaluation metric without requiring class-imbalance correction techniques.

5.2 Age Distribution



Most patients in the dataset fall between 45 and 65 years of age, which aligns with the expected real-world age range where cardiovascular risk increases significantly.

5.3 Feature Correlation Heatmap



Features such as Chest pain type, Exercise angina, ST depression, Slope of ST, and Thallium show comparatively stronger correlation with the target variable (Heart Disease), consistent with established medical risk indicators for heart disease.

6. Model Used

To ensure a robust and well-supported conclusion, three supervised classification algorithms were trained and compared rather than relying on a single model:

- **Logistic Regression:** A linear baseline model that estimates the probability of heart disease using a sigmoid function over a weighted combination of input features. Chosen for its simplicity, speed, and interpretability.
- **Support Vector Machine (SVM):** Used with an RBF (radial basis function) kernel to capture non-linear decision boundaries between the disease and non-disease classes.
- **Random Forest:** An ensemble of 100 decision trees, chosen for its robustness to noise, ability to model non-linear feature interactions, and resistance to overfitting compared to a single decision tree.

7. Training Process

- The cleaned dataset (270 records, 13 features) was split into training (216 records, 80%) and testing (54 records, 20%) sets using stratified sampling to preserve class balance.
- Feature values were standardized using StandardScaler, fitted only on the training data and then applied to the test data, to avoid data leakage.
- Each of the three models (Logistic Regression, SVM, Random Forest) was trained independently on the same scaled training set using a fixed random seed (random_state=42) for reproducibility.
- After training, each model generated predictions on the unseen 20% test set, which were then evaluated using standard classification metrics.

8. Evaluation Metrics and Results

Since this is a classification problem, the models were evaluated using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

Metric	Logistic Regression	SVM	Random Forest
Accuracy	0.8519	0.8148	0.8148
Precision	0.7857	0.7692	0.7692
Recall	0.9167	0.8333	0.8333
F1-Score	0.8462	0.8000	0.8000

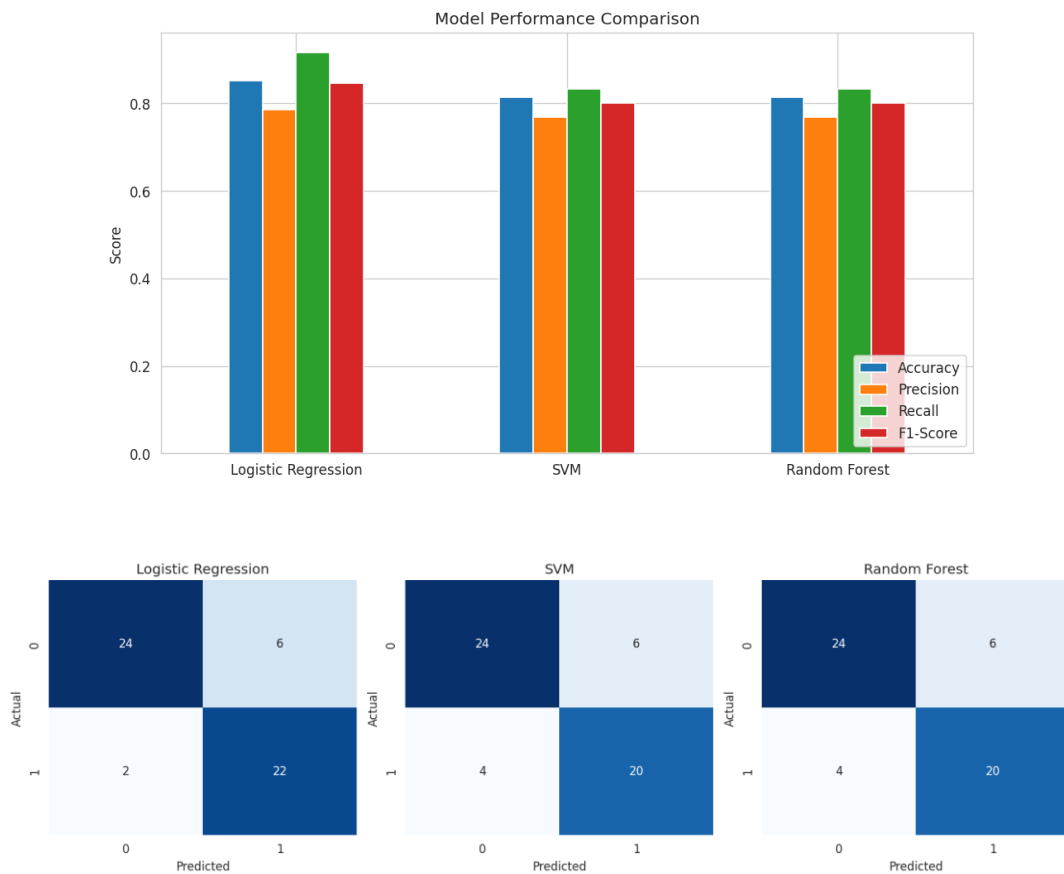


Figure 5: Confusion matrices for Logistic Regression, SVM, and Random Forest (left to right).

Analysis and Interpretation

Logistic Regression achieved the best overall performance among the three models, with the highest accuracy (85.19%) and the highest F1-score (0.8462). It also achieved the highest recall (91.67%), meaning it correctly identified the most actual heart disease cases, missing only 2 out of 24 positive cases in the test set. SVM and Random Forest produced identical results on this test split, both achieving 81.48% accuracy, 76.92% precision, 83.33% recall, and an F1-score of 0.80. This is a plausible outcome on a relatively small test set of only 54 samples, where different models can converge to the same predictions, especially on a clean and well-structured dataset like this one. Across all three models, recall consistently exceeded precision, indicating that the models are slightly biased toward predicting disease presence. This is a desirable property in a medical screening context, since a false negative (missing an actual disease case) is generally more costly than a false positive.

9. Conclusion

This project successfully demonstrates the complete supervised machine learning workflow applied to a real-world healthcare problem — heart disease prediction. Starting from problem identification and dataset acquisition, through data preprocessing, exploratory data analysis, model training, and rigorous evaluation, the project followed each stage of a standard ML pipeline.

Among the three classification models compared, **Logistic Regression delivered the strongest overall performance**, with the highest accuracy, recall, and F1-score. SVM and Random Forest performed identically on this test split and offered a reasonable alternative, though slightly behind Logistic Regression on all metrics. This suggests that for this particular dataset, the relationship between the clinical features and the presence of heart disease is largely well captured by a linear decision boundary.

Such predictive models can act as a useful preliminary screening aid for healthcare professionals, helping flag patients who may require further diagnostic testing. However, this model is intended purely as an academic and exploratory exercise and is not a substitute for professional medical diagnosis.

Future Scope: Performance could be further improved through hyperparameter tuning (e.g., GridSearchCV), exploring additional ensemble methods such as Gradient Boosting or XGBoost, and validating the model on a larger and more diverse patient population to improve generalizability.

10. References

1. Heart Disease Prediction Dataset — Kaggle ([link](#))
2. Scikit-learn Documentation. scikit-learn.org/stable
3. Pandas Documentation. pandas.pydata.org/docs
4. Seaborn Documentation. seaborn.pydata.org
5. Matplotlib Documentation. matplotlib.org/stable/contents.html